

RUSHIL GUPTA

Sector 85, Mohali, Punjab, India | 8375911611 | rushilgupta4@gmail.com | rushilgupta.in

EDUCATION

Ashoka University

B.Sc (Hons) in Mathematics and Computer Science with Research

Sonapat, Haryana

Expected 2027

- CGPA: 3.91/4, Major CGPA: 4.0/4.0
- Relevant Coursework: Fourier Series, Matrix Analysis, Metric and Topological Spaces, Advanced Algorithms, Reinforcement Learning, Machine Learning, Cryptography, Stochastic Calculus and Mathematical Finance
- Teaching Assistantships: Reinforcement Learning, Introduction to Machine Learning, Numerical Algorithms and Optimisation, Introduction to Computer Science

RESEARCH EXPERIENCE

Undergraduate Researcher

Aug 2024 – Present

Safexpress Centre for Data, Learning and Decision Sciences

- Developed a provably efficient estimator for simulating stochastic differential equations under a fixed budget.
- Analyzed the sample complexity of diffusion models for accurately modeling reweighted distributions.
- Developed methods that use small agent-based models to approximate the spread of disease in larger models.

Research Intern

Feb 2024 – Dec 2024

KOITA Centre for Digital Health Ashoka

- Developed infrastructure for a cohort study integrating data streams from multiple wearables and a custom app.
- Worked on an LLM solution to automate informed consent, leveraging agentic models for improved performance.

PUBLICATIONS

Fundamental limits for weighted empirical approximations of exponentially tilted distributions Jan 2026

AISTATS 2026 (<https://arxiv.org/abs/2512.23979>)

- Provided scaling limits for empirical reweighted estimators of reweighted distributions, finding regimes where the estimator is accurate.

Ensemble model with inherently good quality data improves early mortality prediction in ICUs Jan 2025

Critical Care Medicine

- Developed a method to identify high-quality subsets of ICU data; 10% improvement in early mortality prediction.

WORK EXPERIENCE

Platform Teams Intern

Nov 2024 – Jan 2025

SARVAM AI

Bengaluru, India

- Designed and implemented a versioning system for voice agents for incremental updates and seamless deployment.
- Implemented the first unit and integration tests for backend microservices, improving API reliability.

POSITIONS OF RESPONSIBILITY

Head of Events

Sep 2024 – May 2025

Ashoka Computer Science Society

- Led a team of 20 members in planning and executing events, including talks, workshops, and mixers.
- Initiated Leetcode Saturdays; hosted weekly competitive programming sessions for students to solve problems.

Co-Founder and Head of Events & Logistics

Mar 2024 – Dec 2024

Ashoka Health Research Club

- Ideated and co-founded the club, being involved in the selection process of the core and events team.
- Established a framework connecting students with faculty research, offering observership and internship tracks.

TECHNICAL SKILLS

Languages: Python, JavaScript, Julia, Scala, OCaml, HTML, CSS

Frameworks: PyTorch, FastAPI, Django, Pandas, NextJs, React Native, LangChain

Tools: Docker, Git, CI/CD, AWS, GCP, Firebase, PostgreSQL, Redis